Executive Summary

Simultaneous Group-Envelope Bounds for Γ-Robust Multiple-Choice Knapsack Problems

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Many resource-allocation decisions choose exactly one option per group. Protecting their shared constraint with a Bertsimas–Sim uncertainty budget produces a robust multiple-choice knapsack problem. The classical threshold reduction is polynomial, but a large set of distinct deviations can make complete relaxation certification expensive.

After dualizing the capacity constraint of a fixed-threshold MCKP, all threshold-dependent baselines cancel. Between consecutive deviations in a group, the remaining envelope is the maximum of one constant and one common-slope line. Prefix and suffix maxima with range accumulation evaluate one multiplier over all B thresholds in $O(B + K \log(B + 1))$ time and $O(B + K)$ working storage, rather than $\Theta(BK)$ direct score evaluations, for K options.

Weak duality makes the interval minimax value a valid upper bound on every feasible fixed-threshold LP in the interval; strong LP duality gives equality on feasible singleton intervals. A new epigraph-dual mapping proves that the exact minimax envelope bound is no larger than the bounded-threshold group-clique LP. A best-first splitting algorithm retains lower and upper bounds and certifies the maximum LP value over the complete threshold family to a requested scaled tolerance.

The internally protocol-fixed gates were serialized before the final confirmatory rerun. Across 60 primary instances, the compressed certificate reaches scaled tolerance 10^{-6} throughout, wins all 60 paired timings, and is 2.46 times faster in geometric mean than a sparse bounded-threshold group-clique LP (95% design-stratified bootstrap interval 2.29–2.65). Speedup increases from 1.75 at 360 groups to 3.80 at 1,440 groups. All 36 robustness cases reach tolerance; an eight-instance stress panel reaches 5,760 groups with a 5.77-fold geometric-mean advantage. A post-confirmatory UCI-calibrated panel wins 9/9 timings with a 1.80-fold advantage. Independent algebraic, LP, and exhaustive-scan checks find maximum error 6.58×10^{-8}.

The released implementation also embeds either interval bound in an exact integer threshold search. A separate audit confirms objective agreement and valid gap accounting, while showing that integer subproblem work can erase the oracle advantage and that compact SCIP remains preferable on many-breakpoints instances. The analysis therefore concerns simultaneous all-threshold Lagrangian evaluation for exactly-one groups, not the classical threshold reduction, a new robust formulation, or universal integer-solver superiority.